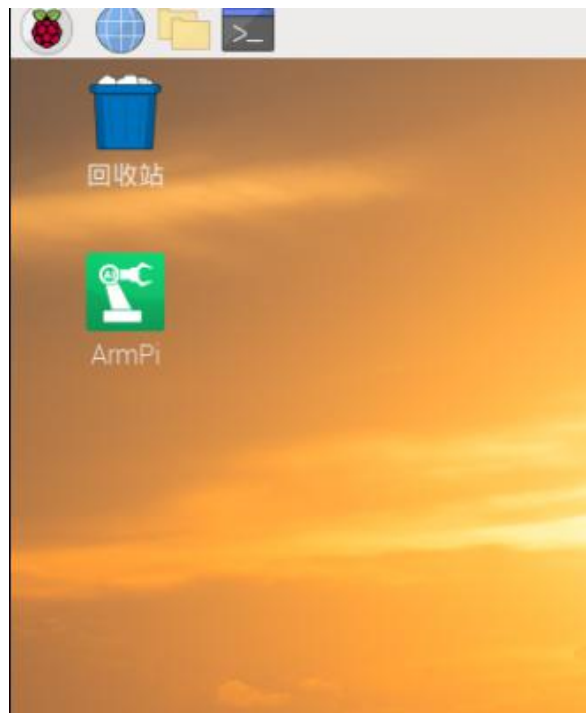


System Introduction

1. Desktop Instruction

After remote connection via VNC, the Raspberry Pi system desktop is as shown in the figure below:



: ArmPi PC software which includes servo control, servo debugging and camera debugging.



: LX terminal is used to input instructions to operate.

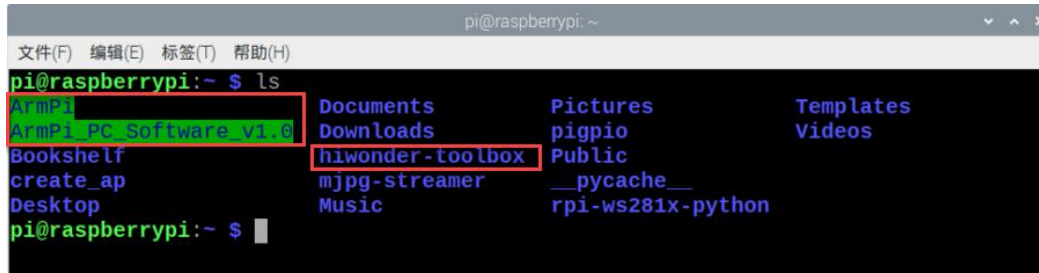


: File manager is a visual file directory interface. (Recommended for beginner users who are not familiar with the command line)

Tip: For more detailed instructions of PC software, please go to folder “4.Advanced Program Lesson/2. Action Programming”.

2. Programming Instruction

- 1) Click  icon or press “Ctrl+Alt+T” to open LX terminal.
- 2) Enter “ls” command and then press “Enter” to list all the documents. As shown in the figure below:



```

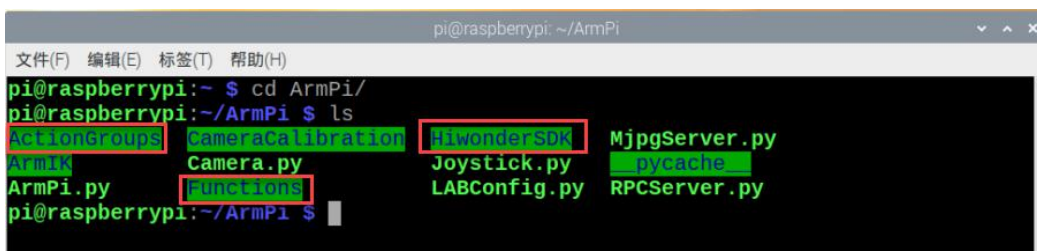
pi@raspberrypi: ~
文件(F) 编辑(E) 标签(T) 帮助(H)
pi@raspberrypi:~ $ ls
ArmPi          Documents      Pictures       Templates
ArmPi_PC_Software_v1.0  Downloads     pigpio         Videos
Bookshelf      hiwonder-toolbox  Public
create_ap      mjpg-streamer  __pycache__
Desktop        Music          rpi-ws281x-python
pi@raspberrypi:~ $

```

Directory	Function
ArmPi	Store all the games and related program source code
ArmPi_PC_Software_v1.0	ArmPi PC software source code (based on Qt)
hiwonder-toolbox	Wi-Fi management tool

Note: For AI vision games, you only need to check the folder “ArmPi”.

- 3) Enter “cd ArmPi/” to open all the games and program source code.
- 4) In the ArmPi, enter “ls” command in the following three main directories. As shown below:



```

pi@raspberrypi: ~
文件(F) 编辑(E) 标签(T) 帮助(H)
pi@raspberrypi:~ $ cd ArmPi/
pi@raspberrypi:~/ArmPi $ ls
ActionGroups  CameraCalibration  HiwonderSDK  MjpgServer.py
ArmIK         Camera.py          Joystick.py  __pycache__
ArmPi.py      functions          LABConfig.py RPCServer.py
pi@raspberrypi:~/ArmPi $

```

Directory	Function
Functions	The directory where the AI vision game program is located.
ActionGroups	Storage path of action groups (used to save actions edited by PC software).
HiwonderSDK	Underlying file path (for hardware control)

5) Enter the "CD functions /" and "LS" instructions in turn again. Let's take a look at the corresponding games of the program:

Program Name	Game
Calibrating.py	Calibrate position
ColorTracking.py	Color tracking
ColorSorting.py	Color sorting
ColorPalletizing.py	Intelligent stacking